

LAB INSTRUCTION SHEET: LAB EXERCISE 04

Topic: Vector Space Modeling — Bag of Words (BoW) & Cosine Similarity

Course Title:	Natural Language Processing	Student Name:	_____
Course Code:	CS-602 / DS-604	Roll Number:	_____
Lab Duration:	02 Hours	Date Performed:	____ / ____ / 2026

1. OBJECTIVE & LEARNING OUTCOMES

The objective of this laboratory exercise is to provide hands-on experience in representing textual data as numerical vectors and computing pairwise document similarity. By the end of this session, students will be able to:

- Understand the mechanics of tokenization, vocabulary extraction, and term frequency representation using the **Bag of Words (BoW)** model.
- Implement term feature extraction using Python's `scikit-learn` library (`CountVectorizer`).
- Compute geometric distance and angle between document vectors using **Cosine Similarity**.
- Construct a basic information retrieval / search query matching mechanism.

2. BACKGROUND & MATHEMATICAL FORMULATION

2.1 Bag of Words (BoW) Model

The Bag of Words model converts text into fixed-length numerical vectors by counting word occurrences, ignoring grammar and word order. Given a corpus of documents $D = \{d_1, d_2, \dots, d_m\}$ and an extracted vocabulary of unique tokens $V = \{t_1, t_2, \dots, t_n\}$, document d_i is represented as a vector $v_i = [f_{i1}, f_{i2}, \dots, f_{in}]$, where f_{ij} is the count of term t_j in document d_i .

2.2 Cosine Similarity

Cosine similarity measures the similarity between two non-zero vectors by calculating the cosine of the angle between them in a multidimensional vector space. For two document vectors A and B :

$$\text{Cosine Similarity}(A, B) = \cos(\theta) = (A \cdot B) / (\|A\| \|B\|) = \sum A_i B_i / (\sqrt{\sum A_i^2} \times \sqrt{\sum B_i^2})$$

The output ranges from **0.0** (completely orthogonal / no overlapping terms) to **1.0** (identical term proportions).

3. ENVIRONMENT SETUP & SOFTWARE INSTRUCTIONS

Students must perform this lab using Python 3.8+ inside a Jupyter Notebook or PyCharm/VS Code environment. Ensure the required libraries are installed before proceeding.

3.1 Package Installation Command

```
pip install numpy pandas scikit-learn
```

3.2 Required Modules Import

```
import pandas as pd
import numpy as np
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

4. LAB TASKS & EXERCISES

Task 1: Bag of Words Matrix Construction (Customer Reviews Dataset)

Scenario: You are given three customer reviews. You need to construct a vocabulary and generate a term-frequency matrix using `CountVectorizer` with English stop words removed.

Sample Input Text:

```
corpus = [
    "The product performance is amazing and fast",
    "The service was fast and performance was great",
    "Terrible customer service and bad performance"
]
```

Instructions:

1. Instantiate `CountVectorizer(stop_words='english')` and fit-transform the corpus.
2. Extract the vocabulary list using `get_feature_names_out()`.
3. Convert the transformed sparse matrix into a Pandas DataFrame for structured presentation.

Task 2: Document Search Engine & Relevance Ranking

Scenario: Build a mini search engine that accepts a search query and ranks documents based on cosine similarity scores.

Sample Input Documents & Query:

```
documents = [
    "Machine learning algorithms analyze structured data effectively",
    "Deep learning and neural networks excel at processing unstructured data",
    "Natural language processing helps computers understand human language",
    "Python is widely used for machine learning and data science"
]

query = ["machine learning algorithms for data"]
```

Instructions:

1. Fit `CountVectorizer` on `documents`.

2. Transform both `documents` and `query` into numerical vector arrays.
3. Compute pairwise cosine similarity using `cosine_similarity(query_vector, doc_vectors)`.
4. Display ranked documents from highest score to lowest score.

5. LAB VIVA & REFLECTION QUESTIONS

Students must answer the following questions in their lab submission report:

1. **Word Order Invariance:** Why does the sentence *"Dog bites man"* have the exact same Bag of Words representation as *"Man bites dog"*? How does this impact sentiment analysis?
2. **Sparsity Issue:** What happens to the memory size and density of the BoW matrix when the corpus contains 100,000 unique vocabulary words?
3. **Zero Similarity:** Explain why Document 3 in Task 2 receives a Cosine Similarity score of `0.0000` when queried against *"machine learning algorithms for data"*.

6. SUBMISSION INSTRUCTIONS (GITHUB)

Follow these steps to submit your lab assignment via GitHub:

1. **Repository Setup:** Create a public GitHub repository named `nlp-lab-04-bow-cosine`.
2. **Repository Structure:**
 - `solution.py` or `solution.ipynb` — Executable code for Task 1 and Task 2.
 - `README.md` — Contains screenshot outputs of both tasks and answers to the Viva/Reflection questions in Section 5.
3. **Commit & Push:** Commit your files with a clear message (e.g., `feat: complete lab 04 tasks and viva answers`) and push to the `main` branch.
4. **LMS Submission:** Submit the public URL of your GitHub repository on the department LMS portal before the due date.